

Leo Marcinkus

Project 2

ECEN 3400

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Rail Gun – Project 2

Background/Part 1:

The magnetic rail gun is a potentially promising machine in the development of projectile propulsion. It was first seriously developed in the early 20th century with a notable small model created in 1917. Once energy conversion and storage technology modernized, the U.S. Navy and various large weapons contractors started to invest in large scale models. This predominately took place in the early 2000s when large-scale railgun prototypes were designed to fit on naval ships due to their long range and incredible projectile launch velocity. Although seeming promising, the development of the railgun proved challenging with constant issues having to be worked around despite its relatively straightforward design.

The magnetic rail gun consists of two parallel conductive rails to form a circuit with a moving armature between them. The projectile can be the armature itself, or an object placed in front of the armature so as they slide together, the object gets launched with great speed. To actually propel the armature, a large power source drives current through the rails and armature. This in combination with the magnetic field it generates creates what is known as a Lorentz force. That being a force applied to a particle due to electric and magnetic fields. This drives the armature between the rails and out a long barrel at a great speed.

The circuit and physics are simple. The rails and armature create one side of a loop. From the large power source explained earlier, electrons travel up the positive rail, across the armature, and then back through the negative rail. This generates a magnetic field around the rails and a magnetic flux that generates a massive force directed down the rails. Although simple, the design is not the most sustainable. Due to the incredibly large currents, usually up to megaamperes in military weapons, thermals are a major concern. The hotter the temperature, the more stress that will be put on the mechanical parts as well. Due to the great speeds, there is a large amount of pressure between the rails and armature. This means that each time the rail gun is fired, physical damage wears the rails drastically decreasing the lifetime of the machine. The other major challenge is converting that massive electrical energy into kinetic energy. It is possible, as it is the only way the machine functions, but it requires incredibly robust electrical discharging systems. This is usually in the form of large capacitor banks as short but powerful bursts are needed and are the advantages of the rail gun.

Although the railgun can deliver excellent performance and high energy output at long range, there remains a large gap between small scale demonstrations and true largescale use of the machine. The downsides are significant at larger scales, especially when trying to be implemented into current systems. After recently trying to implement these extraordinary machines, the U.S. Navy analyzed that in order to be integrated onto current and newer naval vessels, an additional power management system will be required to prevent interference with other systems. As of recent, the U.S. military has suspended further heavy development of the rail gun due to all these drawbacks and instead invest in other weapon systems.

However, if these engineering difficulties can be overcome, rail guns can be revolutionary. Current long-range weapons rely on multiple external systems to help them

function such as the ballistic missile. Not only is it interception a major vulnerability with missiles, but they are also extraordinarily expensive to operate. Whereas rail gun launched projectiles get shot at such an incredible velocity, it is virtually impossible to intercept. The massive energy output can pierce heavy armor and can change current countermeasures. The quick fire rate and low cost to operate could change modern warfare and strategy. A relatively simple circuit and physics can reinvent military capabilities if its technical difficulties can be hurdled.

A diagram illustrating the most basic concepts is below:

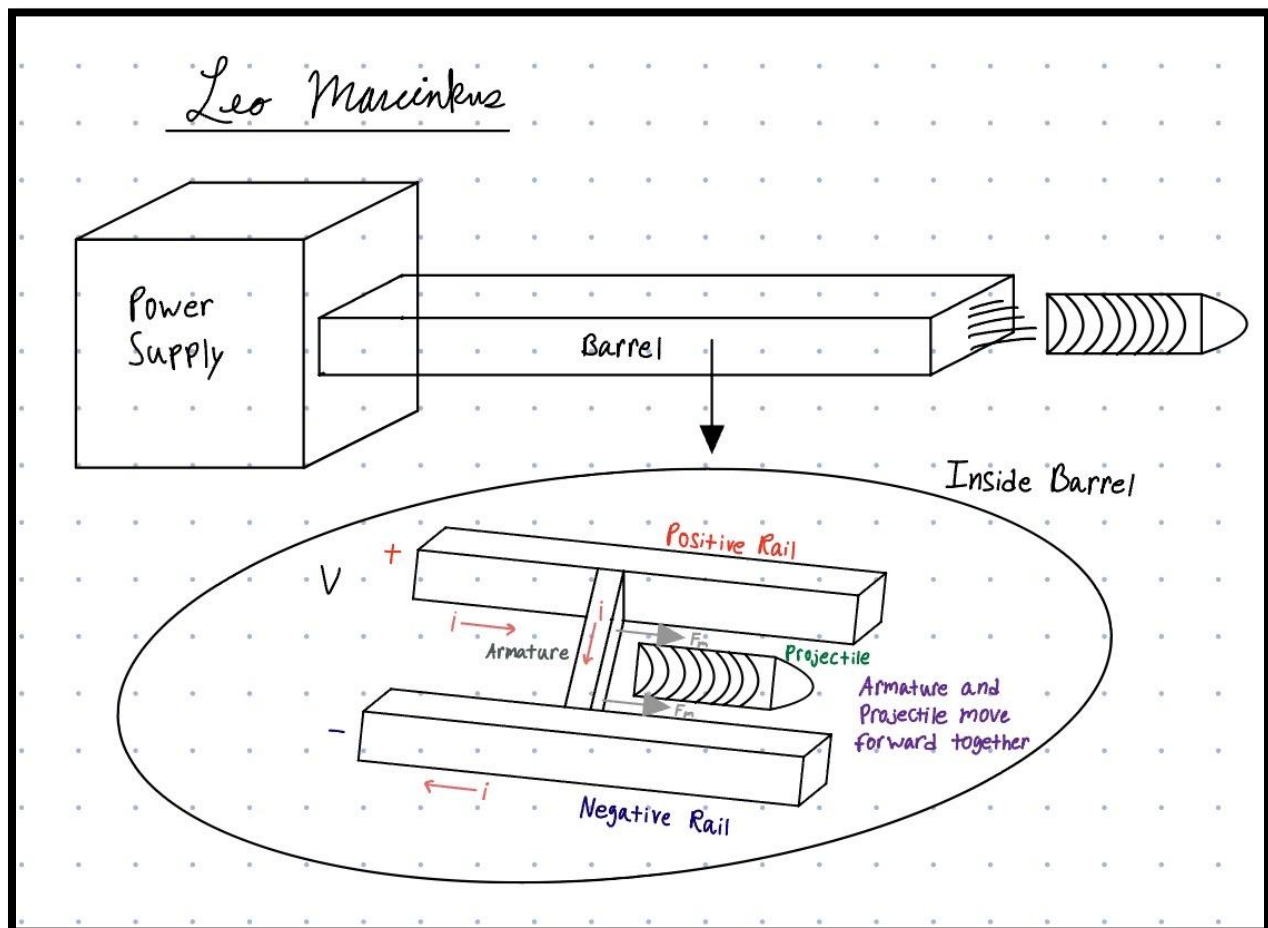


Figure 1: Basic Rail Gun Diagram

Part 2:

The magnetic rail propulsion system I designed can be assembled with simple components and utilizes Lorentz force. To create this force, two square copper rails and a copper armature are connected to a capacitor bank. The rails are mounted to a wooden board with clamps. And the armature overhangs the rails on the top with two additional copper plates to ensure stability. The current from the capacitors (16V, 1500F) flows through the first copper rail, then the armature, and then through the second rail. This is what creates the force to launch the armature forwards, in turn sending the car with it. Optionally, an additional wooden board can be mounted above the rails to ensure a more stable path and that nothing goes flying upwards. The estimated total cost comes out to be ~\$1,443. The most expensive parts being the super capacitors and protective switch.

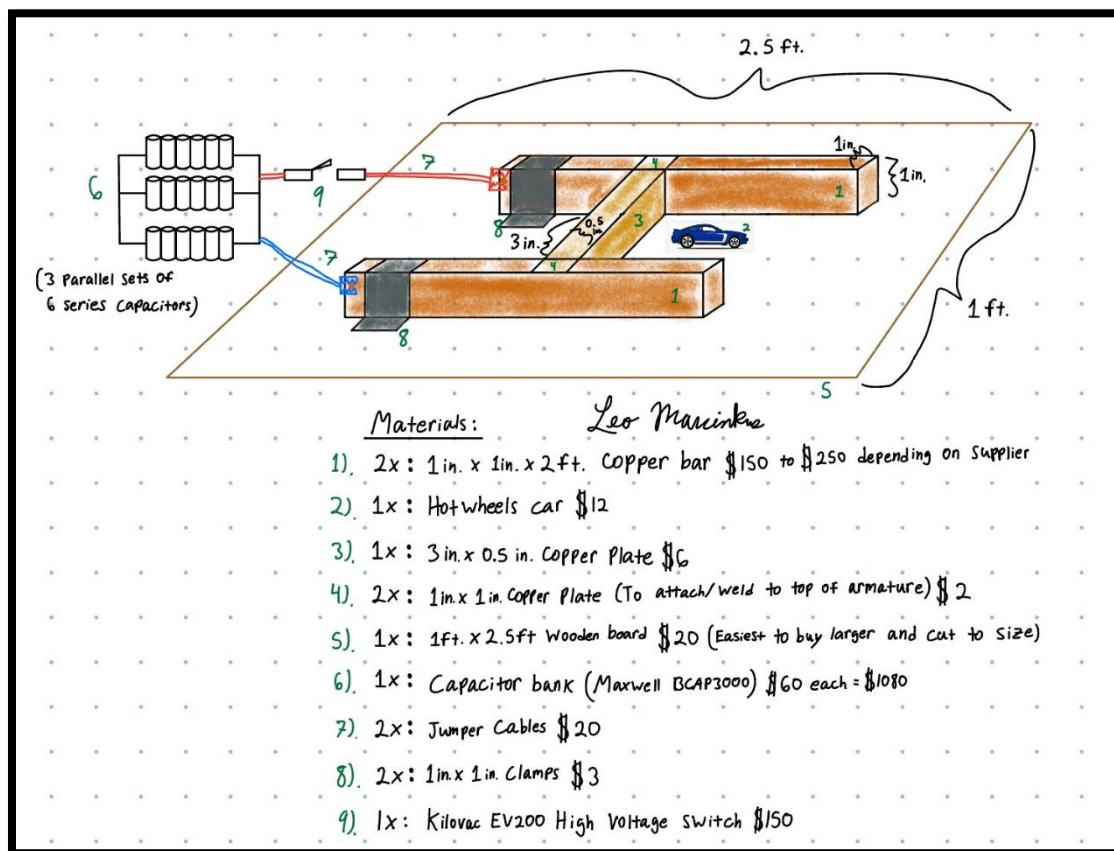


Figure 2: Design and Materials

Calculations:

Assuming:

- Rail length (d) = 2ft = 0.6096 m
- Rail separation (s) = 3 in = 0.0762 m
- Armature length (l) = 3 in = 0.0762 m
- Rail cross-section (A) = 1 in x 1 in = 6.45 x 10⁻⁴ m²
- Copper resistivity = 1.68 x 10⁻⁸ Ωm
- Mass of car = 50 g = 0.05 kg
- Armature mass is negligible
- Copper rails are acting as ideal wires
- Constant magnetic field between the rails
- Capacitor Bank: V = 16.2 V F = 1500F, R (including switch) = 0.88 mΩ

Setup:

$$I = \frac{V}{R} = \frac{16.2V}{0.00088\Omega} \approx 18,409.091 \text{ A}$$

$$B = \frac{\mu_0 I}{2\pi s} = \frac{(4\pi \times 10^{-7})(18,409.091)}{2\pi(0.0762)} \approx 0.0483T = 48.3 \text{ mT}$$

$$F_{EM} = \frac{\mu_0}{2\pi} \times \frac{I^2 \ell}{s} = \frac{(4\pi \times 10^{-7})}{2\pi} \times \frac{(18,409.091)^2(0.0762)}{(0.0762)} \approx 67.78 \text{ N}$$

$$F_{friction} = \mu mg = 0.53 \times 0.05 \times 9.81 = 0.26 \text{ N}$$

Velocity:

$$\tau = RC = 0.00088 \times 1500 \approx 1.32s$$

$$a(t) = We^{\frac{-2t}{\tau}} - \frac{F_{friction}}{m}$$

$$v(t) = \int We^{\frac{-2t}{\tau}} - \frac{F_{friction}}{m} = \frac{W\tau}{2} \left(1 - e^{\frac{-2t}{\tau}}\right) - \frac{F_{friction}}{m} t$$

$$x(t) = \int v(t) = \frac{W\tau}{2} \left(t - \frac{\tau}{2} (e^{\frac{-2t}{\tau}})\right) - \frac{F_{friction}}{2m} t^2$$

$$k = \frac{\mu_0}{2\pi} = 2 \times 10^{-7}$$

$$W = \frac{kI^2}{m} = \frac{(2 \times 10^{-7})(18,409.091)^2}{0.05} \approx 1355.58 m/s^2$$

$$\frac{F_{friction}}{m} \approx 5.199 m/s^2$$

$$v(t) = \frac{(1355.58)(1.32)}{2} \left(1 - e^{\frac{-2t}{(1.32)}}\right) - 5.199t$$

$$x(t) = \frac{(1355.58)(1.32)}{2} \left(t - \frac{(1.32)}{2} \left(e^{\frac{-2t}{(1.32)}}\right)\right) - 2.5995t^2 = 0.6096m \rightarrow t_{exit} \approx 0.0303s$$

$$v_{exit} = \frac{(1355.58)(1.32)}{2} \left(1 - e^{\frac{-2(0.0303)}{(1.32)}}\right) - 5.199(0.0303) \approx 40 m/s$$

The car should exit with an approximate exit velocity of 40m/s.

Energy/Power:

$$E = VIt = (16.2 \text{ V})(18409.091)(0.0303 \text{ s}) \approx 9036.286 = 9.0362 \text{ kJ}$$

$$P_{average} = \frac{E}{t} = \frac{9.0362 \text{ kJ}}{0.0303 \text{ s}} \approx 298.227 \text{ kW}$$

Conclusion:

This design is capable of launching the car out at a speed of 40 m/s, just shy of 90 miles per hour. However, this design is also extremely dangerous as it lacks any major safety features. It assumes the capacitors are safely charged and will work as intended. There is no fuse included or leakage prevention. Since all the magnetic energy is coming purely from the rails, I needed an extreme amount of current without the help of other magnets. That's why I decided on a power bank filled with those extremely high capacitance ultracapacitors. I was originally going to use a car battery but quickly realized the armature would not be able to overcome the friction force of 0.26 N. I also suspect that this design will destroy itself after one launch. The system requires an incredible energy need of 9.0362 kJ. Although it might work once, this is not the most feasible design.

This project has been extremely useful in learning about energy demands and how demanding magnetic forces can be. Future improvements would include more robust materials and the inclusion of more safety features beyond just one switch. Although just a design project, I believe this explains the incredible power of electromagnetic forces and what applications they can be applied to.

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